

Pay Transparency and Gender Differences in Job Search

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September 13, 2026

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Abstract

This paper evaluates the impact of the pay transparency law on labor outcomes. We use an equilibrium model with subjective beliefs to illustrate how wage beliefs affect job search behavior and outcomes. Under the assumptions that women have lower wage beliefs than comparable men and that more wage information in job postings reduces the gender gap in wage beliefs, the model predicts that the pay transparency law will decrease the gender wage gap. Using CPS data and a difference-in-differences design, we find that the law modestly reduced overall wages—driven primarily by a decline in male wages among employer stayers. Consistent with the model, the gender wage gap narrowed sharply among employer changers, falling from 17.5% to 3.5%. This decline was primarily driven by rising female wages; male wages also fell modestly, though the decline was not statistically significant. Despite a noticeable effect on the gender wage gap for job changers, pay transparency did not affect the gender gap in job mobility, employment, or labor force participation.

Keywords: pay transparency, gender wage gap, job search, salary history bans

JEL: J16, J31, J63, J78

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1 Introduction

In recent years, pay transparency laws have gained traction globally as a policy tool to address wage inequality and improve labor market outcomes. In the United States, several jurisdictions have enacted laws requiring employers to disclose wages in job postings. Colorado was the first state to mandate comprehensive wage transparency in job advertisements, effective January 1, 2021. This legislation requires employers to include wage ranges and benefits in all job postings. Other states, including New York, California, and Washington, have since followed suit with similar mandates.

Pay transparency laws can provide workers with new information about expected wages. For example, in Colorado, the share of job postings with salary information was below 10% prior to 2021 but rose to approximately 60% after the law’s implementation (Feng, 2025). Research shows that workers often have misperceptions about the wages they could earn at other firms (Jäger et al., 2024), with women, in particular, tending to underestimate their potential earnings compared to men (Cortés et al., 2023). Given the widespread misperceptions workers have about wages, providing wage information could help correct these inaccurate beliefs, especially for women, encouraging them to pursue higher-paying job opportunities. Despite the theoretical benefits, however, empirical evidence on the impact of pay transparency on labor market outcomes is limited.

This study aims to fill this gap by providing a comprehensive assessment of the effects of Colorado’s pay transparency law on labor market outcomes, focusing on gender differences. We use an equilibrium model with subjective wage beliefs, which illustrates how these beliefs influence job search behavior and labor market outcomes. The model assumes that women tend to have lower wage beliefs than comparable men and that more wage information in job postings reduces the gender gap in wage beliefs. Based on these assumptions, the model predicts that the pay transparency law will decrease the gender wage gap. Using data from the Current Population Survey (CPS) and a difference-in-differences (DID) design, we find that the law modestly reduced overall wages, primarily driven by a decline in male wages among employer stayers. Consistent with the model, the gender wage gap narrowed remarkably among employer changers, falling from 17.5% to 3.5%. This decline was mainly driven by rising female wages, although male wages also fell modestly (though the decline was not statistically significant). While the law had a noticeable effect on the gender wage gap for employer changers, it did not influence the gender gap in job mobility, employment, or labor force participation.

To guide the empirical analysis, we use an equilibrium model with subjective wage beliefs, building upon Jäger et al. (2024). We make two assumptions to connect the model predictions

to the empirics: (1) Women have lower beliefs about their expected wages than comparable men (Cortés et al., 2023; Kiessling et al., 2019; Roussille, 2024); (2) The pay transparency law decreases the gender gap in wage beliefs. In the model, female workers accept lower wage offers and sort into low-wage sectors due to their lower wage beliefs. This sorting behavior is consistent with existing literature (e.g., Blau and Kahn (2017); Card et al. (2016)). As wage beliefs converge across genders, women’s reservation wages increase, making them less likely to accept low-wage offers. This reduces their sorting into lower-paying jobs, which in turn narrows the gender wage gap.

We employ a DID approach to empirically assess the impact of Colorado’s pay transparency law. We compare labor market outcomes in Colorado to those in a control group of states that implemented statewide salary history bans (SHBs) before 2021 but did not adopt pay transparency mandates between 2021 and 2023. SHB states are an appropriate control group because they, like Colorado, enacted labor market regulations to promote pay equity. The DID approach isolates the impact of Colorado’s policy from national trends that could affect both the treated and control states. To examine gender-specific effects and test model predictions, we extend the DID approach by interacting the treatment indicator with a female dummy to examine differential effects by gender, comparing gender gaps in labor outcomes in Colorado before and after the law, relative to those in the control states.

A key methodological challenge arises from the staggered adoption of SHB laws across control states. Since SHBs may also affect labor outcomes, pooling all SHB states using a common time window could confound the effects of SHBs with those of Colorado’s policy. To address this, we group control states by SHB adoption year and estimate separate DID models for each group, restricting the sample window to start after the latest SHB implementation. We then pool the results across groups, weighting each group-specific estimate by its share of total observations.

We find that Colorado’s pay transparency law led to a modest but statistically significant decline in overall wages, primarily driven by wage reductions among workers who stayed with the same employer. Since the law aims to promote pay equity, we next focus on its gender-specific impacts, particularly changes in the gender wage gap. Among workers who changed employers, the effects are striking. In the most comprehensive specification, which controls for occupation and industry fixed effects, the gender wage gap in Colorado fell from 17.5% to just 3.5%, an 80% reduction. Notably, the gender wage gap declined more in female-dominated occupations, where employer compliance with wage disclosure was higher. These findings are consistent with theoretical predictions that pay transparency can reduce gender wage disparities.

In contrast, the gender wage gap did not narrow significantly for workers who stayed with

the same employer, even among those who switched occupations internally. While the law requires pay ranges in both internal and external postings, its effects appear concentrated among external movers. Additionally, there is suggestive evidence of a decline in men’s wages—approximately 3% for employer stayers and up to 8% for employer changers. Although the estimated decline in male wages is imprecise, it aligns with prior research showing that transparency can reduce male wages by making employers more cautious about raises that might trigger equity concerns (Baker et al., 2023; Cullen and Pakzad-Hurson, 2023). The results on the gender wage gap suggest that while pay transparency can substantially reduce the gender wage gap by improving outcomes for women who change employers, it has limited impact on incumbent workers—highlighting both the promise and the boundaries of what such policies can achieve.

We also examine whether the law affected broader gender gaps in job mobility, employment, or labor force participation by estimating difference-in-differences models that allow for gender-specific treatment effects. The estimated coefficients are small and statistically insignificant, indicating that the policy did not meaningfully change these aggregate outcomes for women relative to men.

This study contributes to the growing literature on pay transparency policies and their effects on workers’ outcomes, particularly gender pay disparities. Prior research distinguishes between horizontal transparency (sharing pay information within a firm), vertical transparency (revealing pay tied to career progression), and more recently, cross-firm transparency (publicly posting wages in job ads). Horizontal transparency, such as when employees learn coworkers’ salaries, has been shown to reduce unexplained pay gaps, improve perceptions of fairness, and prompt job search or negotiation among lower-paid groups, particularly women (Card et al., 2012; Gamage et al., 2020; Obloj and Zenger, 2022; Blundell, 2021; Lyons and Zhang, 2023; Baker et al., 2023). Complementary evidence from Denmark and the UK reinforces the potential of transparency policies to promote pay equity: in Denmark, firm-level gender pay gap reporting curtailed top-end male wage growth (Bennedsen et al., 2022), while UK employers responded to disclosure mandates with modest adjustments to reduce disparities (Duchini et al., 2020). However, this form of transparency may lead firms to compress wage distributions in response to perceived inequities, potentially reducing incentives for top performers (Cullen and Pakzad-Hurson, 2023). Vertical transparency—making pay levels at higher ranks visible—can increase work effort (Cullen and Perez-Truglia, 2022).

This study explores a less studied form of pay transparency: cross-firm pay transparency, which involves revealing the pay information of other firms. Two recent studies examine pay disclosure mandates similar in spirit to Colorado’s and assess their impact on the gender wage gap. Skoda (2022) studies a 2018 Slovakian policy requiring employers to disclose the fixed

component of wages in job advertisements. Using linked job board and administrative data, he finds a modest wage increase (around 3%) for new hires at firms that complied with the mandate. However, the gender wage gap persisted, likely because only base pay was disclosed while variable compensation remained opaque and negotiable—conditions that may limit women’s bargaining power. [Frimmel et al. \(2022\)](#) analyze a 2011 Austrian law mandating the inclusion of minimum wage offers in job ads. They find no effect on average wages but report a narrowing of the gender wage gap in vacancies requiring urgent hiring, consistent with increased bargaining leverage for women in those contexts. While both studies provide useful evidence on transparency and gender inequality, they focus on partial or minimum-pay disclosure and rely on simple pre-post comparisons without a credible control group. Our study contributes to this literature by evaluating a comprehensive pay transparency law that introduces a new and stronger form of disclosure—requiring employers to post the full salary range in all job advertisements, rather than just base pay or minimum wage offers, as well as banning the ask of salary history. Using a difference-in-differences design with SHB states as a control group, and a two-step estimation approach that accounts for staggered adoption timing, we provide more credible evidence that full pay transparency substantially reduces—and nearly eliminates—the gender wage gap among employer changers.

Second, this paper also contributes to the literature on gender differences in job search behavior, wage expectations, and negotiation—factors that help explain persistent gender wage gaps. Women tend to set lower reservation wages, accept offers more quickly, and negotiate less frequently than men, partly due to greater risk aversion and lower expectations ([Biasi and Sarsons, 2022](#); [Cortés et al., 2023](#); [Miano, 2023](#)). These differences contribute to persistent pay disparities. However, recent work shows that access to reliable wage information can close these gaps. [Yilmaz and Brandl \(2024\)](#) use a lab-in-the-field experiment with Austrian students and job seekers to test how different employer practices in job advertisements affect gender pay expectations. They compare scenarios with salary range disclosure to those with vague terms like “salary negotiable”. Their findings show that only clear salary range disclosure significantly reduces gender gaps in wage expectations. This evidence underscores that mandatory salary range disclosure is a particularly powerful tool for promoting pay transparency and reducing gender disparities in wage expectations. [Roussille \(2024\)](#) find that providing the median market wage data on a job search platform eliminates gender differences in ask salaries, and [Leibbrandt and List \(2015\)](#) demonstrate that stating “salary negotiable” in job ads removes the gender gap in negotiation initiation in experiment settings. Our study extends this line of work by examining the effects of a real-world, large-scale policy mandating full salary range disclosure in job postings. We show that the law substantially reduces the gender wage gap among workers who change employers but has no meaningful effect

on incumbent workers—revealing both the promise and the boundaries of pay transparency policies. Although our study cannot directly test the mechanisms involved, the observed wage effects are consistent with the idea that transparency corrects wage misperceptions, encourages women to move to higher-paying jobs, and/or increases women’s bargaining power.

A third strand of literature examines the important role of job mobility in driving wage growth and labor market dynamics. Workers who switch employers tend to experience faster wage increases by escaping firm-level wage compression and using outside offers for better pay (Topel, 1991; Haltiwanger et al., 2022; Moscarini and Postel-Vinay, 2019). However, women face structural barriers such as caregiving responsibilities, occupational segregation, limited networks, and employer biases that restrict their job mobility and bargaining power, leading to smaller wage gains from job changes compared to men (Goldin, 2014; Bouchard et al., 2023; Manning, 2003). Our study contributes to this literature by showing that the gains from job mobility depend not only on workers’ ability and willingness to move, but also on the broader information environment in which mobility occurs. While we do not find evidence that the policy increased overall or female job mobility, we show that women who do change employers experience substantial wage gains. The evidence suggests that women, in particular, will benefit from clearer wage information when access to outside options improves, even if their rates of mobility remain unchanged. This is consistent with prior research emphasizing the role of outside options in women’s wage growth (Goldin, 2014; Haltiwanger et al., 2022; Moscarini and Postel-Vinay, 2019).

The rest of the paper will proceed as follows. Section 2 will discuss the institutional details of the Colorado pay transparency law. Section 3 will introduce the CPS data. Section 4 will lay out an equilibrium model with subjective beliefs. Section 5 will report the results on labor outcomes. Section 6 will present robustness checks and section 7 will conclude.

2 Institutional Background

2.1 Colorado’s Equal Pay for Equal Work Act

On January 1, 2021, Colorado’s Equal Pay for Equal Work Act (EPEWA) went into effect. This act applies to all employers and employees in Colorado, both public and private. This act requires employers to (1) provide the wage rate or range and employment benefits in “both external job postings (e.g., public advertisements) and internal job notices (e.g., company-wide emails)” (Colorado Department of Labor and Employment, 2024), (2) keep records of job descriptions and wage rate history for each employee, and (3) notify employees of promotional opportunities. It also prohibits employers from asking about or relying on a job applicant’s

salary history, i.e., the salary history ban. Our focus will be on the first component of this act: including wage information in job postings.

Specifically, this act requires the disclosure of “the hourly or salary compensation, or a range of the hourly or salary compensation, and a general description of all the benefits and other compensation to be offered to the hired applicant” ([Colorado General Assembly, 2019](#)). The salary range must be for the particular job advertised and may extend from the lowest to the highest amount the employer genuinely believes it would pay for that position ([Colorado Department of Labor and Employment, 2021](#)). For example, the Colorado Department of Labor states that “an employer cannot post a \$70,000-\$100,000 range for a junior accountant position just because it pays senior accountants at the higher end of that range.” This act does not allow employers to post a salary range with no lower or upper bound, like \$30,000 and above or below \$80,000. An employer is permitted to pay more or less than the indicated range when a vacancy is eventually filled.

Enforcement of the law came mainly through education and rarely financial penalties. Individuals can submit compliant forms to the Colorado Department of Labor if they find an employer violates the law. After investigating a complaint, the state labor department would first issue a no-fines Warning and Order to the employer if it found a violation such that the employer had the chance to comply. Most employers complied after learning of a violation and thus avoided a fine ([Chuang, 2022](#)). According to the law, the non-compliance-related fine can range from \$500 to \$10,000 per violation.

To provide context on the policy’s implementation, the companion paper [Feng \(2025\)](#) examines the first-stage effects of Colorado’s pay transparency law—specifically, how employers responded to the wage disclosure requirement. Using online job postings from LinkUp, [Feng \(2025\)](#) documents a substantial increase in compliance: the share of job postings in Colorado that included salary information rose by over 40 percentage points following the mandate, with increases observed across all occupations. The rise in wage posting was especially pronounced in occupations with a higher share of female workers. These results demonstrate a strong first-stage response and lay the groundwork for the current paper’s focus on downstream labor market effects.

2.2 Salary History Bans

Salary history bans (SHBs) had emerged across the United States as a legislative effort to promote pay equity and reduce wage discrimination before Colorado implemented its pay transparency law. Oregon was the first state to enact a statewide SHB in October 2017. Following Oregon, several other states have enacted statewide SHBs, including Alabama,

California, Connecticut, Delaware, Illinois, Maine, Massachusetts, New Jersey, New York, Oregon, Washington, and Vermont, with laws taking effect between 2017 and late 2020. Several counties and cities have also implemented SHBs at the local level.¹ Like the SHB component in Colorado’s pay transparency law, these statewide SHBs typically prohibit employers from asking job applicants about their prior salary history and from using that information to set compensation. However, none of these prohibitions prevent prospective employees from voluntarily disclosing prior salary information without being prompted.² Key components of these SHBs often include restrictions on salary history inquiries during the hiring process at any stage and, in some cases, requirements to provide salary ranges for open positions upon request. The primary policy goals of these prohibitions are to prevent the perpetuation of historical wage disparities-particularly those affecting women and minority groups - by ensuring that compensation offers are based on the applicant’s qualifications and the requirements of the job, rather than on prior earnings that may reflect past discrimination.

Studies examining the effects of SHBs find significant effects for both employers and employees. For employers, SHBs increase the likelihood of including wage information in job postings by 2-3 percentage points (Bessen et al., 2024; Sran et al., 2020). For workers, these bans reduce the gender wage gap by 1-2% in the overall population (Hansen and McNichols, 2020; Sinha, 2022) and by up to 7.9% among those who change employers (Bessen et al., 2024). These findings suggest that SHBs are effective tools for promoting gender pay equity.

3 CPS Data

The dataset is the CPS from 2017 to 2023 (Flood et al., 2023). It complements the job posting data because it is representative of the U.S. population and includes realized wages. The survey lasts four consecutive months, followed by an eight-month break, and then another four consecutive months before participants leave the survey. In the monthly basic files, the survey records demographic information for each household member, such as age, sex, education, and location. It also records their labor market outcomes, such as employment status, industry, and occupation. For six of the eight months of the survey, employed participants are asked if they changed employers from the previous month. The monthly basic files do not contain data on individual earnings. The Outgoing Rotation Group supplement records earnings information in the 4th and 8th months of the survey. For those workers paid an hourly wage, CPS reports how much the respondent earned per hour in the current job, and we use the

¹For a running list of states and localities that have implemented any sort of salary history bans, please visit <https://www.hrdive.com/news/salary-history-ban-states-list/516662/>.

²In some states, even if the applicant voluntarily discloses prior salary information, these SHBs do not permit the use of salary history to set compensation.

reported hourly wage rate as the outcome variable. For salaried workers, we compute the hourly rate by dividing the weekly earnings by the reported usual number of hours per week the respondent reports being at their main job.³

We restrict our sample to the civilian non-institutionalized population aged between 22 and 64. We focus on this particular group of individuals because they typically have enough time to graduate from college and start working by age 22 and remain in the labor force until retirement.

4 An Illustrative Equilibrium Model with Beliefs

In this section, we establish a simple equilibrium model with subjective beliefs to illustrate how (inaccurate) beliefs about wages can affect job search behavior and outcomes. Our model builds upon [Jäger et al. \(2024\)](#).

4.1 Setup

Environment There are N firms that have homogenous production functions in the labor market that set and post their wages. Each firm offers the same wage w to all its workers. There are L risk-neutral individuals. Initially, individuals draw a job search cost c , where $c \sim \mathcal{N}(\mu, \sigma^2)$. After realizing their search cost, individuals decide whether to stay in the labor force and search for jobs or to drop out of the labor force permanently. This decision is irreversible.

Unemployed individuals who choose to stay in the labor force incur the search cost c and receive a random job offer from a firm. They learn about the wage w and can then decide whether to accept the job offer. For simplicity, this model does not include on-the-job search. Employed workers leave for unemployment at an exogenous job destruction rate s . [Figure A1](#) illustrates this decision process. The figure shows how individuals draw search costs, make participation decisions, receive offers conditional on paying the cost, and transition between employment and unemployment.

Workers, Beliefs, and Job Search There are two types of individuals: a share α of type A individuals and a share $1 - \alpha$ of type L individuals. Type A ($\tau = A$) individuals and type L individuals ($\tau = L$) differ only in their beliefs about the highest wage $\tilde{w}_\tau^{\max}$:

$$\tilde{w}_L^{\max} = w^{\max} - \delta \tag{1}$$

³This method for calculating the hourly wage of salaried workers aligns with [Lachowska et al. \(2022\)](#).

$$\tilde{w}_A^{\max} = w^{\max}. \quad (2)$$

Type A workers have accurate beliefs about the highest wage, while type L workers have downward biased beliefs about the highest wage. We do not allow workers to learn and update their beliefs by changing employers, which seems unrealistic. However, this is not problematic because pay transparency can affect incumbent workers' beliefs without changing their employers first, and belief updating by changing employers is not the relevant mechanism in this context.

Unemployed workers accept a job offer w_j from firm j if

$$w_j \geq \tilde{w}^{\max}. \quad (3)$$

Given that on-the-job search is not possible in this model, workers are almost “locked” at the job offer they accept for sufficiently low job destruction rate s and sufficiently large probability of receiving a job offer providing at least $\tilde{w}^{\max}$. This rationalizes the decision rule that workers will not accept job offers with wages lower than $\tilde{w}^{\max}$.

Thinking backwards, given the realized search cost c , an individual will choose to enter the labor force and search for jobs only if

$$\tilde{w}^{\max} \geq c.$$

Suppose the cumulative distribution function of the search cost c is $F(c)$, then the labor force participation rate for type A worker is

$$F(\tilde{w}_A^{\max}) = F(w^{\max}),$$

and for type L worker is

$$F(\tilde{w}_L^{\max}) = F(w^{\max} - \delta).$$

Firms and Wage Setting Firms produce a homogeneous good using a decreasing-returns production function $f(l) = l(w)^\eta$, where $\eta \in (0, 1)$. A firm's employment $l(w_j|w_{-j})$ depends on the wage it pays along with those paid by other firms. Given its own wage w_j and the external wage structure of other firms w_{-j} , firm j 's profit is

$$\pi(w_j|w_{-j}) = l(w_j|w_{-j})^\eta - w_j \cdot l(w_j|w_{-j}). \quad (4)$$

4.2 Single-Wage Equilibrium

In this model, there exists a competitive single-wage equilibrium where the wage is equal the marginal product of labor. The competitive wage, represented by w^* , is defined as $w^* = \eta(l^*)^{\eta-1}$. In a profit-maximizing environment, firms have no incentive to pay more than this competitive wage.

Labor market clearing pins down the equilibrium firm size l^* :

$$N \cdot l^* = (1 - u)L(\alpha F(w^*) + (1 - \alpha)F(w^* - \delta))$$

where $u = \frac{1}{1+s}$ is the unemployment rate.

$$l^* = (\alpha F(w^*) + (1 - \alpha)F(w^* - \delta)) \frac{L}{(1 + s)N}.$$

4.3 Conditions for Single-Wage Equilibrium

The single-wage equilibrium sustains only when there is no profitable unilateral deviation from offering the competitive wage, w^* . If a profitable deviation exists, the only possibility is that deviants pay a lower wage such that they could only recruit type L workers, not type A workers. In this case, deviants would offer the reservation wage of type L workers:

$$w' = \tilde{w}_L^{max} = w^* - \delta.$$

Since a deviant keeps only type L workers, its employment is:

$$l(w') = F(w') \frac{(1 - \alpha)L}{(1 + s)N}.$$

Deviant's profit would then be:

$$\pi(w') = \left(F(w') \frac{(1 - \alpha)L}{(1 + s)N} \right)^\eta - w' F(w') \frac{(1 - \alpha)L}{(1 + s)N}.$$

Deviation is *not profitable* if $\pi(w') < \pi(w^*)$:

$$\delta < \frac{Y^\eta - w^*Y - X^\eta + w^*X}{X}, \tag{5}$$

where

$$X = F(w') \frac{(1 - \alpha)L}{(1 + s)N}$$

and

$$Y = \frac{L}{(1+s)N} ((1-\alpha)F(w') + \alpha F(w^*)).$$

Equation 5 implies that other things being equal, the greater the δ , the more likely deviation is to be profitable.

4.4 Segmented (Two-Wage) Equilibrium

When type L workers have a large downward bias in beliefs about the highest wage (a large δ) and Equation 5 does not hold, it is profitable for some firms to deviate from offering w^* and mark down the wage by δ , which results in a two-wage equilibrium with a high wage (w_h) and a low-wage (w_l) sector. In equilibrium, the share of high-wage firms, denoted by β , adjusts so that low-wage and high-wage firms are equally profitable. The belief gap, the degree of bias δ , supports this segmentation.

Separation Flows The separation rate for any type of worker at any firms is just the job destruction rate s .

Unemployment Rates Balancing the outflows from unemployment (LHS) and inflows (RHS) for type A and type L workers respectively:

$$\beta u^A F(w_h) \alpha L = s(1 - u^A) F(w_h) \alpha L$$

$$u^L F(w_l) (1 - \alpha) L = s(1 - u^L) F(w_l) (1 - \alpha),$$

which gives:

$$u^L = \frac{s}{1+s}$$

$$u^A = \frac{s}{2\beta + s - 1}.$$

Recruiting Flows High-wage firms get both types of workers from the unemployment pool. The recruiting flow for each type of worker at high-wage firms are:

$$R_h^A = s u^A F(w_h) \alpha \frac{L}{N}$$

$$R_h^L = s u^L F(w_l) (1 - \alpha) \frac{L}{N}$$

Low-wage firms get only type L workers from the unemployment pool:

$$R_l^A = 0, \quad R_l^L = s u^L F(w_l) (1 - \alpha) \frac{L}{N}$$

Firm Size Firm j has a separation rate s_j^τ and a recruiting flow R_j^τ for worker type τ . Thus, in steady state, the flows of separation and recruitment are in balance:

$$sl_j^\tau = R_j^\tau,$$

which gives

$$l_l^A = 0, \quad l_l^L = \frac{1}{1+s} F(w_l)(1-\alpha) \frac{L}{N},$$

$$l_h^A = \frac{2\beta-1}{\beta(2\beta+s-1)} F(w_h) \alpha \frac{L}{N}, \quad l_h^L = \frac{1}{1+s} F(w_l)(1-\alpha) \frac{L}{N}.$$

Low-wage firms keep only type L workers. The employment of each low-wage firm is:

$$l_l = l_l^A + l_l^L$$

The employment of each high-wage firm is:

$$l_h = l_h^A + l_h^L$$

The Wage in the High-Wage Sector Within the high-wage sector, the same logic applies as is in the single-wage equilibrium. The wage in this sector, w_h , is equal to the competitive wage:

$$w_h = \eta (l_h)^{\eta-1}. \quad (6)$$

The Wage in the Low-Wage Sector Low-wage firms keep only type L workers so this sector's wage w_l is equal to the reservation wage of type L workers, $w_h - \delta$:

$$w_l = w_h - \delta. \quad (7)$$

In equilibrium, low-wage firms and high-wage firms are equally profitable: $\pi(w_l) = \pi(w_h)$.

In the segmented equilibrium, type L workers' average wage ($\bar{w}_L$) is lower than type A workers' average wage ($\bar{w}_A$) because type L workers sort into low-wage firms while type A workers sort into high-wage firms. The wage gap between type A and type L workers is

Wage Gap

$$\bar{w}_A - \bar{w}_L = w_h - (\beta w_h + (1-\beta)w_l) = (1-\beta)\delta. \quad (8)$$

In the segmented equilibrium, the labor force participation rate of type L workers (H_L) is lower than that of type A workers (H_A), because type L workers perceive labor force participation as less profitable than type A workers. However, the employment rate of type L

workers $(1 - u_L)$ is higher than that of type A workers $(1 - u_A)$ because type L workers have a lower reservation wage. The labor force participation gap and the employment gap between type A and type L workers are

Labor Force Participation Gap

$$H_A - H_L = F(w_h) - F(w_l) \quad (9)$$

Employment Gap

$$(1 - u_A) - (1 - u_L) = \frac{1}{1 + s} - \frac{2\beta - 1}{2\beta + s - 1} \quad (10)$$

4.5 Implications and Limitations

As δ , the belief gap decreases, the model produces the following predictions, also illustrated in [Figure B4](#):

1. The wage gap between type A and type L workers $(\bar{w}_A - \bar{w}_L)$ will decrease;
2. The labor force participation gap between type A and type L workers $(H_A - H_L)$ will decrease;
3. The employment gap between type A and type L workers $((1 - u_A) - (1 - u_L))$ will increase.

To connect the model implications to the empirics, we need to make two assumptions:

1. Women have lower beliefs about their expected wages than comparable men ([Cortés et al., 2023](#); [Kiessling et al., 2019](#); [Roussille, 2024](#)).

All else equal, women are more likely to be a type L worker relative to men. For illustrative purposes, this section considers an extreme scenario where all type L workers are women and all type A workers are men. In reality, there is considerable individual-level heterogeneity in the accuracy of wage beliefs ([Jäger et al., 2024](#); [Jiang and Zen, 2023](#)), and men do not necessarily have accurate beliefs about wages.

2. The pay transparency law decreases the gender gap in wage beliefs δ .

By providing expected wage information in job postings, pay transparency laws can reduce the gender belief gap in expected wages (δ) . However, this can only be assumed as we do not observe wage beliefs.

Gender Belief Gap and Sorting The model demonstrates that female workers tend to sort into low-wage sectors due to their lower beliefs about expected wages. This sorting behavior is consistent with existing literature (e.g., [Blau and Kahn \(2017\)](#); [Card et al. \(2016\)](#)).⁴ According to the model, a reduction in δ increases women’s reservation wages relative to men, making women less likely to accept low-wage offers. As shown in [Figure B4](#), a smaller gender belief gap (δ) decreases the gender wage gap by increasing the proportion of women working in high-wage sectors (β), thus mitigating the trend of women sorting into lower-paying jobs.

Employment This model predicts that women’s employment will decrease relative to men as a result of decreased gender belief gap δ . The rationale is that women raise their reservation wages, which decreases the share of low-wage firms that only recruits women.

Labor Force Participation The model also predicts that women’s labor force participation will increase relative to men due to a decreased gender belief gap δ . The underlying logic is that as women perceive higher returns from job searching, they are more likely to participate in the labor force.

Gender Belief Gap and Bargaining In addition to sorting into lower-paying jobs, research also identifies another pathway that contributes to the gender wage gap: a bargaining channel in which women receive or negotiate for lower wages compared to men (e.g., [Bertrand \(2011\)](#); [Biasi and Sarsons \(2022\)](#); [Card et al. \(2016\)](#); [Roussille \(2024\)](#)). A limitation of this model is that it does not feature wage bargaining. However, pay transparency may influence the bargaining channel, leading women to demand higher wages during negotiations. Both experimental and observational evidence shows that although women demand a lower amount in bargaining, giving women information about the amount demanded by others in a similar bargaining situation eliminates the gender bargaining gap ([Rigdon, 2012](#); [Roussille, 2024](#)).

5 Research Design and Empirical Specification

To estimate the effect of Colorado’s pay transparency law, we employ a DID framework that compares labor market outcomes in Colorado—where the law took effect in 2021—to outcomes in a set of control states that did not adopt such a law. This approach allows us to isolate the effect of the policy from broader national trends affecting wages, employment, and job mobility.

The control group consists of states that implemented statewide salary history bans (SHBs) before 2021 and did not introduce additional pay transparency mandates during the post-treatment period. These include Alabama, Connecticut, Delaware, Illinois, Maine, Massachusetts, Oregon, and Vermont. [Table A1](#) lists the effective dates of SHBs in each of

⁴In the CPS data, we also observe that on average, women work in lower-wage occupations than men.

these states. We exclude states like California and Washington, which implemented their own transparency mandates in 2023, and New Jersey and New York, which initially applied SHBs only to public-sector employers. SHB states offer a useful comparison group because, like Colorado, they passed laws aimed at improving pay equity and protecting workers. This makes them more similar in spirit to Colorado than states with no such policies,

A key challenge arises from the staggered timing of SHB adoption across control states. Because SHBs themselves can influence labor market outcomes—such as wage offers and mobility—pooling all SHB states with a common pre-period could bias the estimated treatment effect. For instance, [Sinha \(2022\)](#) finds that SHBs reduce the gender wage gap and weaken the link between prior and future wages.

To address this issue, we group SHB states by the year their bans took effect and estimate separate DID models for each group. For each group, we restrict the sample window to start in the month of the latest SHB implementation among the included control states. This ensures that all control states maintain a stable SHB policy regime throughout the estimation period. For example, in the 2019 group—Connecticut, Illinois, Alabama, and Maine—the sample begins in October 2019. Colorado is included in each group-specific comparison, but is considered “treated” only starting in January 2021, when its pay transparency law took effect. Groupings are as follows: Oregon and Delaware (2017), Massachusetts and Vermont (2018), and Connecticut, Illinois, Alabama, and Maine (2019). We estimate separate DID models for each group and compute a pooled effect by weighting each group-specific estimate by its share of total observations.

This grouped comparison approach is similar in spirit to the staggered adoption framework of [Callaway and Sant’Anna \(2021\)](#), but adapted for a setting where differential timing exists in the control group, as opposed to the treatment group. It also helps address concerns about small sample sizes that would arise in state-by-state comparisons.

The baseline DID specification is as follows:

$$Y_{ismt} = \beta \cdot \text{Treat}_{ismt} + X_{ismt}\Gamma + \lambda_s + \lambda_t + \lambda_m + \theta_{c(i)} + \epsilon_{ismt}, \quad (11)$$

where Y_{ismt} is the log hourly wage of individual i in state s , calendar month m , and year t . To improve statistical power, the data are aggregated to the annual level due to limited monthly sample sizes in the CPS Outgoing Rotation Group. The treatment indicator Treat_{ismt} equals one for individuals in Colorado observed after the law’s implementation in 2021. The vector X_{ismt} includes controls for gender, age, race, education, occupation, industry, public/private sector, full-/part-time status, and whether paid hourly. The model includes state (λ_s), year (λ_t), and month (λ_m) fixed effects., as well as occupation and sector fixed effects ($\theta_{c(i)}$).

To examine dynamic effects and test for pre-trend violations, we estimate the following event-study version of the DID model:

$$Y_{ismt} = \sum_{t=-4, t \neq -1}^1 \beta_t \cdot \mathbf{1}_t \cdot \text{Treat}_{ismt} + X_{ismt}\Gamma + \lambda_s + \lambda_t + \lambda_m + \theta_{c(i)} + \epsilon_{ismt}. \quad (12)$$

Here, $\mathbf{1}_t$ indicates the number of years relative to the policy implementation in 2021, with $t = -1$ as the omitted reference year.

To assess how the law affected gender disparities in labor market outcomes, we extend the analysis by interacting the DID treatment indicator with a female dummy. This approach estimates whether the policy had differential effects by gender, comparing changes between women and men in Colorado relative to those in the control states:

$$Y_{ismt} = \beta \cdot (\text{Female}_i \times \text{Post}_t \times \text{CO}_s) + \alpha \cdot (\text{Post}_t \times \text{CO}_s) + \delta_1 \cdot (\text{Female}_i \times \text{CO}_s) + \delta_2 \cdot (\text{Female}_i \times \text{Post}_t) + \delta_3 \cdot \text{Female}_i + X_{ismt}\Gamma + \lambda_s + \lambda_t + \theta_{c(i)} + \epsilon_{ismt}. \quad (13)$$

The dependent variable Y_{ismt} may represent: (1) log hourly wage, (2) a binary indicator for having changed employer, (3) a binary indicator for employment, or (4) a binary indicator for labor force participation. The coefficient β captures the change in the gender gap in Colorado relative to control states after the law took effect.

Finally, we estimate an event-study version of the above model to explore the evolution of gender gaps over time:

$$Y_{ismt} = \sum_{t=-4, t \neq -1}^1 \beta_t \cdot \mathbf{1}_t \cdot (\text{Female}_i \times \text{Post}_t \times \text{CO}_s) + \sum_{t=-4, t \neq -1}^1 \alpha_t \cdot \mathbf{1}_t \cdot (\text{Post}_t \times \text{CO}_s) + \delta_1 \cdot (\text{Female}_i \times \text{CO}_s) + \delta_2 \cdot (\text{Female}_i \times \text{Post}_t) + \delta_3 \cdot \text{Female}_i + X_{ismt}\Gamma + \lambda_s + \lambda_t + \theta_{c(i)} + \lambda_s \times \text{Female}_i + \epsilon_{ismt} \quad (14)$$

This specification introduces state-by-gender fixed effects to flexibly control for gender-specific baseline differences across states.

6 The Effect of Pay Transparency on Labor Outcomes

6.1 Effect on Wages

[Table A2](#) reports the DID estimates of the impact of Colorado's pay transparency law on wages. Across the full sample (columns 1-3), the estimated effect of the law is negative and

statistically significant—about a 1.6% to 2.0% decline in log hourly wages, significant at the 5% level.

The event-study plot (Panel A of [Figure A2](#)) provides further context. While there are some pre-treatment deviations, the post-treatment estimates show a negative effect that persists through the post-period.

Next, we examine treatment effect heterogeneity by job mobility status. We define employer changers as employed individuals who changed employers before their wage was reported in months 4 or 8 of the CPS survey; all others are classified as employer stayers.⁵ We focus on employer changers because they are more likely to respond to pay transparency. Workers seeking new jobs in the external labor market face greater uncertainty about pay, and public wage postings can help them identify better-paying opportunities and negotiate more effectively. The estimates for employer changers (columns 4–6) are more positive than those for the full sample, ranging from -0.19% to 0.15%, but are imprecisely estimated and statistically insignificant. For employer stayers (columns 7–9), the estimates show statistically significant negative effects—about a 1.4% to 2.1% decline in wages, with the largest effect (column 9) significant at the 5% level. Overall, the pay transparency law had a modest but statistically significant negative effect on wages, particularly for workers who remained with their current employers. However, to fully understand the implications of this finding, it is important to decompose the effect by gender, as the overall decline may mask differential impacts on men and women.

6.2 Effect on Gender Wage Gap

Colorado’s pay transparency law is intended to promote pay equity and reduce the gender wage gap. According to the prediction from the job search model in [section 4](#), increased wage information should reduce the gender gap in wage beliefs and, consequently, narrow the gender wage gap. In this section, we test the model’s prediction and assess whether the law achieves its intended goal.

First, we present the time-varying effects of Colorado’s pay transparency law on the gender wage gap. As shown in Panel B of [Figure A2](#), the event study estimates fluctuate over time. Prior to 2021, the gender wage gap in Colorado and SHB states follows a similar trend, with the exception of 2019. After the law’s enactment, the estimated effect is significantly positive only in 2022 and close to zero in other post-treatment years.

[Table A3](#) presents the difference-in-differences estimates from [Equation 13](#). The coefficient of interest, $Female \times Post \times CO$, captures the change in the gender wage gap attributable

⁵Since employer change information is only collected for six of the eight survey months, some changers may be misclassified as stayers.

to the law. A positive coefficient implies a narrowing of the gap. $Post \times CO$ captures the change in male wages, and $Female$ represents the baseline gender wage gap in SHB states.

Columns 1-3 report estimates for the full sample. The gender wage gap appears to narrow by 2.6% to 3.1%, with columns 1 and 3 showing statistical significance at the 10% level. The law’s estimated effect on male wages ($Post \times CO$) is negative and significant in these specifications, ranging from 2.9% to 3.3% declines.

Columns 4-6 present results for employer changers. Among this subgroup, the gender wage gap narrows more substantially—by 14.1% to 15.9%, depending on the specification. In the most saturated model (column 6), which includes occupation and industry fixed effects, the gap declines by 15.9%, significant at the 5% level. The $Female \times CO$ interaction term indicates that, prior to the law, the gender wage gap among employer changers was larger in Colorado than in the SHB states, with estimated differences ranging from 7.1% to 9.7% (coefficients between -0.071 and -0.097 for $Female \times CO$). This pattern may reflect the earlier adoption of SHBs in the control (SHB) states. The magnitude of this pre-existing gap is also consistent with [Bessen et al. \(2024\)](#), who find that SHBs reduced the gender gap by 7.9% for employer changers using CPS data.

Using coefficients from column 6—the preferred specification—the pre-law gender wage gap in Colorado is given by the sum of the coefficients on $Female$ and $Female \times CO$:

$$\text{Pre-law gap}_{CO} = \beta_{\text{Female}} + \beta_{\text{Female} \times CO} = -0.087 + (-0.088) = -0.175,$$

corresponding to a 17.5% wage gap. The post-law gender gap in Colorado is calculated by adding the coefficients on all relevant interaction terms:

$$\begin{aligned} \text{Post-law gap}_{CO} &= \beta_{\text{Female}} + \beta_{\text{Female} \times CO} + \beta_{\text{Female} \times \text{Post}} + \beta_{\text{Female} \times \text{Post} \times CO} \\ &= -0.087 + (-0.088) + (-0.019) + 0.159 = -0.035, \end{aligned}$$

equivalent to a 3.5% wage gap. This represents an 80% reduction, and the remaining gap is not statistically significant ($p = 0.32$), suggesting that the law is close to eliminated the gender wage disparity for employer changers. Male wages declined by 5.2% to 7.9% following the policy, though these estimates are not statistically significant at the 5% level, likely due to limited sample size and reduced statistical power.

Columns 7–9 report estimates for employer stayers. The coefficients on $Female \times Post \times CO$ range from 2.2% to 2.5%, none of which are statistically significant. In contrast, male wages declined by 2.6% to 2.9% following the policy, with estimates statistically significant at the 5% level. These results imply that, among workers who stayed with the same employer, the transparency law did not raise women’s wages but led to a modest decline in men’s wages,

Since the law requires employers to disclose pay ranges in both *external* and *internal* job postings, we further explore whether women who change occupations within the same employer also benefit. Because CPS does not explicitly track internal job changes, we proxy for this by identifying occupation changes within the same employer.⁶

Table A4 shows that the gender wage gap narrows by 1.3–2.2% for employees who change occupation within the same employer (columns 1–3), but none of these estimates are statistically significant. Occupation stayers show similar changes, if anything even larger—2.4%–3.9% (columns 4–6)—and most of these are also statistically insignificant. Hence, the data do not support meaningful gains for within-employer movers, though measurement error in identifying such moves may attenuate the estimates.

Feng (2025) finds that occupations with a higher share of female workers were more responsive to Colorado’s wage transparency mandate: they saw greater increases in the share of job postings that included salary information (see Figure 8, Panel (b) of Feng (2025)). This suggests that there is much more wage information revealed for female-dominated occupations after the law’s implementation. To examine whether this translated into heterogeneous effect in gender pay equity, we estimate the effect of the law on the gender wage gap separately for occupations above and below 50% female representation.

Table A5 presents the estimates of the law’s effect on the gender wage gap by occupational gender composition. In occupations where women comprise at least 50% of the workforce (columns 1–3), the coefficient on the triple interaction term ($Female \times Post \times CO$) is positive and statistically significant in two of the three specifications, ranging from 0.041 to 0.053. This indicates a decrease in the gender wage gap in these occupations following the law’s implementation. In contrast, in occupations with less than 50% female workers (columns 4–6), the estimates are negative and statistically insignificant, suggesting no detectable impact on the gender wage gap. It is also notable that the baseline gender gap, as captured by the coefficient on *Female*, is smaller in female-majority occupations (about 9–10%) than in male-majority occupations (13–14%). The contrast between the two occupational groups provides suggestive evidence that women benefited more from the wage transparency mandate, consistent with the idea that increased access to salary information can help reduce gender disparities. In male-dominated occupations, the policy resulted in lower wage posting compliance and did not lead to a meaningful reduction in the gender wage gap, suggesting that the policy itself was less effective in these contexts.

In sum, Colorado’s pay transparency law significantly reduced the gender wage gap

⁶For example, the most common within-employer occupation change for women in the sample is from secondary school teacher to elementary and middle school teacher; for men, it is from software developer to web developer.

among workers who changed employers. The gap fell from 17.5% to 3.5%, representing an 80% decline driven by both an increase in women’s wages and a decline in men’s wages. Notably, the gender wage gap declined more in female-dominated occupations, where employer compliance with wage disclosure was higher. The decrease in the gender wage gap aligns with the predictions of the model in [section 4](#): by improving access to wage information, pay transparency raises women’s expectations and facilitates mobility toward better-paying opportunities relative to men. Another possible mechanism is that women’s bargaining power increases with transparency, allowing them to negotiate for and receive higher wages, particularly when changing employers.

In contrast, there is little evidence that the gender wage gap narrows among employer stayers or within-employer movers. This may reflect limitations in the data’s ability to accurately capture internal job transitions, or it could indicate that the wage gains from pay transparency primarily benefit external job seekers. Additionally, although not always statistically significant, male wages declined by roughly 3% among employer stayers and between 5% and 8% among employer changers. This pattern of declining male wages aligns with findings from [Baker et al. \(2023\)](#), who show that salary disclosure policies at Canadian universities reduced the gender wage gap largely through a slowdown in male wage growth. They attribute this effect to two main factors: increased bargaining power among women and employers’ reluctance to raise male salaries in ways that might provoke equity demands. The latter explanation gains some backing from [Cullen and Pakzad-Hurson \(2023\)](#), who argue that pay transparency can suppress overall wages as firms strategically limit raises to avoid triggering costly renegotiations—an effect that could be more pronounced for employees who already earn higher wages.

6.3 Effect on Gender Gaps in Other Outcomes

Job Mobility There is no evidence that Colorado’s pay transparency law affects the gender gap in job mobility. As shown in columns 1–3 of [Table A6](#), the estimated coefficients on $Female \times Post \times CO$ are close to zero and statistically insignificant, indicating that the law did not differentially affect the likelihood of job changes for women relative to men. Men’s probability of changing employers also remains unchanged post-law (row 3). One reason why pay transparency might not affect the gender gap in job mobility could be that wage information in job postings is not sufficiently salient for workers who are not actively seeking new employment and does not significantly alter the beliefs of incumbent workers. It is also possible that job changes depend on multiple factors—such as job satisfaction, benefits, or geographic constraints—beyond just wage information.

Employment and Labor Force Participation We also find no significant effect of the law on gender gaps in employment or labor force participation. Columns 4-7 show that the $Female \times Post \times CO$ interaction terms are statistically insignificant for both outcomes. Although the signs are consistent with the predictions in [section 4](#)—a widening gender employment gap and a narrowing gender LFP gap—the estimated magnitudes are small (less than 0.5 percentage points). This may be because the effects are too small to be picked up, or because the wage information in job postings is not sufficient to change the beliefs of individuals outside the labor force. However, the law is associated with a reduction in average employment and labor force participation of less than 1 percentage point (row 3), which is statistically significant at the 1% level.

Segregation Across Employers We find no economically meaningful effect of the law on the segregation of men and women across employers. [section 4](#) predicts that this segregation should decline: low-wage firms in the segmented equilibrium recruit only workers with downward-biased wage beliefs, and therefore employ a disproportionate share of women, so a convergence in beliefs should reallocate women toward high-wage firms. The CPS cannot test this prediction on sorting, because it records whether a worker changed employers but not which employer. We therefore use LinkedIn employment histories for the same nine states and years, assigning each employer the gender composition of its own workforce before 2021 ([People Data Labs, 2022](#)). [Appendix C](#) describes the construction of the sample. [Table C10](#) shows that men and women are strongly segregated across employers. The average woman is employed at a firm whose workforce is 14.8 percentage points more female than that of the average man (coefficient on $Female$), of which approximately two thirds is attributable to sorting across industries and one third to sorting across firms within an industry. The estimated coefficient on $Female \times Post \times CO$, which represents the treatment effect, is -0.0008 . It presents the sign the model predicts, but the shift is half of one percent of the existing gap between men’s and women’s workplaces. After the law, women in Colorado sorted toward employers with a slightly lower share of women, but the magnitude is too small to support an economic interpretation. One limitation qualifies this estimate. LinkedIn profiles over-represent professional and college-educated workers, so the result bounds the reallocation channel for that population rather than for the CPS sample.

7 Robustness

In this section, we show that our main results are robust to different compositions of the control states and potential Colorado-specific confounders.

7.1 Colorado vs. “Unregulated States”

In the first robustness check, we use an entirely different control group from the main analysis—states that have neither the wage disclosure requirement nor the salary history ban throughout the study period. We refer to these states as “Unregulated States,” which include Arizona, Arkansas, Florida, Idaho, Indiana, Iowa, Kansas, Michigan, Minnesota, Montana, Nebraska, New Hampshire, New Mexico, North Dakota, Oklahoma, Wisconsin, and Wyoming. Compared with SHB states, “Unregulated States” are less similar to Colorado in terms of labor market regulations. However, they still reflect broader national trends in labor market dynamics and may serve as a valid control group, provided the parallel trends assumption holds. This alternative comparison helps assess whether the findings are driven by the specific choice of control states or reflect more general patterns.

[Table A8](#) and [Table A9](#) present the results for wage and non-wage outcomes, respectively. Overall, the findings are consistent with those from the main analysis using SHB states as the control group. Specifically, the estimated reduction in the gender wage gap for employer changers is 9.8% when using Unregulated States as controls, compared to 15.9% with SHB states. Both estimates are imprecise due to the limited sample size for employer changers, and their confidence intervals overlap; the difference between them is not statistically significant. These results suggest that the observed narrowing in the gender wage gap among job changers is robust to alternative control group definitions. Additionally, estimates for non-wage outcomes are close to zero and statistically insignificant, reinforcing the conclusion that the law had limited effects outside of wages.

Importantly, the interpretation of the estimated effects differs slightly across comparison groups. The SHB group provides a policy-relevant benchmark that shares a similar regulatory trajectory with Colorado, while the Unregulated group captures the effect relative to states with no prior SHB or wage-related mandates. However, if the parallel trends assumption holds for either control group—SHB states (after their SHB laws) or Unregulated States—in wage and non-wage outcomes, then estimates using either SHB states or Unregulated States can be interpreted as the causal effect of Colorado’s pay transparency law. Additionally, all estimates for non-wage outcomes remain very close to zero and are statistically insignificant at the 10% level, reinforcing the conclusion that the law’s primary effects are concentrated in the gender wage gap for job changers.

7.2 Within-Colorado DID: Public vs. Private Sector

One concern that is not solved by the previous robustness check, or by any comparison between Colorado and other states, is that effects on labor outcomes may be driven by

Colorado-specific confounders that are not the pay transparency law. To mitigate this concern, we use a within-Colorado comparison and compare the private sector with the public sector. In Colorado, wages for public sector employees had been public information before the pay transparency law, so the pay transparency law should not reveal much new information about wages to the labor market. Therefore, we use the public sector as the control group and the private sector as the treated group and repeat the DID analysis. Based on the classification of workers in the CPS, we define public sector employees as federal, state, or local government employees.

[Table A7](#) display the within-Colorado DID estimates of effects on labor outcomes. Note that wild cluster bootstrap t-statistics, rather than standard errors clustered at the MSA level, are in parentheses, as there are only 7 clusters in the within-Colorado design. Overall, the signs and the magnitude of the estimates are aligned with between-state DID estimates, although the estimates are noisy because of the small sample size. Therefore, it is very unlikely that the effects on the gender gaps in labor outcomes are purely driven by Colorado-specific confounders.

8 Conclusion and Discussion

8.1 Comparison with Other Studies

Ongoing work by [Arnold et al. \(2025\)](#) is the study most closely related to this one. They also examine the effect of Colorado’s pay transparency policy, using self-reported wage data from Glassdoor. Their findings suggest that the law raised realized earnings by 0.9% to 1.4% for the overall sample (including both new hires and incumbents) and by 1.8% to 3% for new hires, with all estimates statistically significant at the 5% level. In contrast, we find a small negative on wages for the full sample and a null effect on the subset of new hires.

These differences may reflect key methodological differences between our studies. First, we use different data sources: they rely on Glassdoor, which covers a selective set of employers and self-reported pay, while we use the nationally representative Current Population Survey. Second, our control groups differ: [Arnold et al. \(2025\)](#) use all other U.S. states as the control group, whereas we restrict the comparison to states that implemented salary history bans before 2021 and did not adopt additional transparency mandates during the post-treatment period.

[Skoda \(2022\)](#) and [Frimmel et al. \(2022\)](#) investigate analogous mandates of posting the minimum wage of the job in Slovakia and Austria respectively. In contrast to Colorado, where substantial noncompliance still exists in late 2022 ([Feng, 2025](#)), Slovakia and Austria have

nearly perfect compliance (80% for Slovakia and nearly 100% for Austria), possibly due to stricter enforcement. [Skoda \(2022\)](#) finds no effect on the gender pay gap in Slovakia, while [Frimmel et al. \(2022\)](#) find a 4.6% decrease in the gender pay gap for urgently needed jobs and no change for non-urgently needed jobs. The reduction in the gender wage gap is driven by women’s improved bargaining power, consistent with our empirical findings in the Colorado context.

[Bessen et al. \(2024\)](#) study the effect of salary history bans in the US, another component of Colorado’s pay transparency law apart from the wage disclosure requirement, and find that salary history bans reduced the gender earning gap for employer changers by 7.9%. Our analysis shows that the overall effect of the pay transparency law for employer changers is 15.9%, which exceeds the effects of its individual components estimated in other contexts-7.9% from the salary history ban and 4.6% from the wage disclosure requirement-and is consistent with the existing literature.

8.2 Alternative Explanations

While we cannot examine changes in wage beliefs directly, none of the alternative explanations account for the full set of empirical results.

One alternative explanation is that pay information in job postings disproportionately increases women’s match efficiency. If the gender wage gap is partly due to women’s poorer match quality, women may benefit more from improved match quality and be more likely than men to move to better job matches and receive higher wages, without necessarily increasing their wage beliefs. However, the employment rate would also increase more for women than for men, since improved match efficiency should also increase the job finding rate of unemployed women more than that of unemployed men. The null effect on the gender difference in employment does not support this explanation.

Another possibility is that pay information in job postings disproportionately reduces women’s (perceived) search costs, especially for high-paying jobs. Women have higher perceived ([Miano, 2023](#)) and actual ([Drahs et al., 2018](#)) search costs than men. If pay transparency reduced women’s search costs more than men’s, it would also reduce the gender wage gap. However, we do not expect pay transparency itself to reduce search costs much more for women than for men. First, pay transparency only reduces the cost of obtaining wage information for specific jobs, not other costs associated with job search, such as transportation costs. Second, it is difficult to argue that pay transparency provides wage information that is new to women but not to men.

8.3 Concluding Remarks

This study provides the first comprehensive evaluation of a pay transparency mandate that combines wage range disclosure with a salary history ban. Using data from CPS, we examine the impact of Colorado’s 2021 law on gender gaps in wages and labor market outcomes. To motivate the analysis, we use a job search model with subjective wage beliefs, which predicts that if women underestimate their market wages relative to men, increased transparency should reduce the gender wage gap by raising women’s wage expectations. The results show that the law led to a 15.9% reduction in the gender wage gap among employer changers in the most saturated specification—reducing the gap from 17.5% to 3.5%, an 80% decline. This narrowing is driven primarily by rising women’s wages; male wages declined by 5.2%, though the change is not statistically significant. Among employer stayers, we find no statistically significant change in the gender wage gap. Notably, the gender wage gap declined more in female-dominated occupations, where employer compliance with wage disclosure was higher. We also find no evidence that the law affected gender differences in job mobility, employment, or labor force participation. These findings indicate that comprehensive pay transparency policies can reduce gender pay disparities, especially by improving outcomes for women who switch employers. However, the effects on workers who remain in their current jobs and on broader labor market behavior appear limited, pointing to both the promise and the limits of transparency-based reforms.

There are several research questions that remain beyond the scope of this study but could be explored in future research. This study relies on observational data and does not directly measure the effect of wage information in job postings on wage beliefs. Understanding how pay transparency affects wage beliefs, particularly the gender belief gap, using direct measures of wage beliefs, would be a valuable direction for future research. Additionally, pay transparency policies alter the wage negotiation process by making expected wages available to job seekers upfront. Investigating how the dynamics of the wage negotiation process between workers and employers evolve in response to pay transparency policies would also be an important area for future study.

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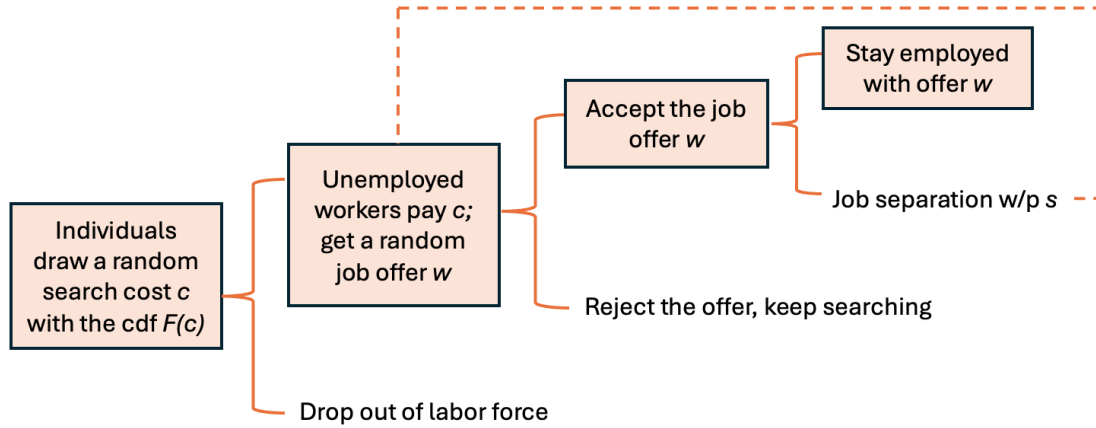
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A Tables and Figures

Figure A1: Job Search Model Structure



Note: This figure illustrates the structure of the job search model. Each period, individuals draw a search cost $c \sim \mathcal{N}(\mu, \sigma^2)$ and decide whether to enter the labor force. Those who participate pay the cost and receive a job offer from a randomly matched firm. They observe the posted wage w and decide whether to accept. Employed workers face an exogenous job destruction rate s and return to unemployment upon separation.

B Additional Tables and Figures

Table A1: Timeline and Legislation Details of Salary History Bans for SHB States

State	Effective Date	Interview Enquiry	Voluntary Disclosure	Ask and Verify after Offer
Oregon	2017-10-06	X	✓	✓
Delaware	2017-12-14	X	✓	✓
California*	2018-01-01	X	✓	✓
Massachusetts	2018-07-01	X	X	✓
Vermont	2018-07-01	X	X	✓
Connecticut	2019-01-01	X	X	✓
Washington*	2019-07-28	X	✓	✓
Alabama	2019-09-01	X	✓	✓
Maine	2019-09-17	X	✓	✓
Illinois	2019-09-29	X	✓	✓
New Jersey*	2020-01-01	X	✓	✓
New York*	2020-01-06	X	✓	✓

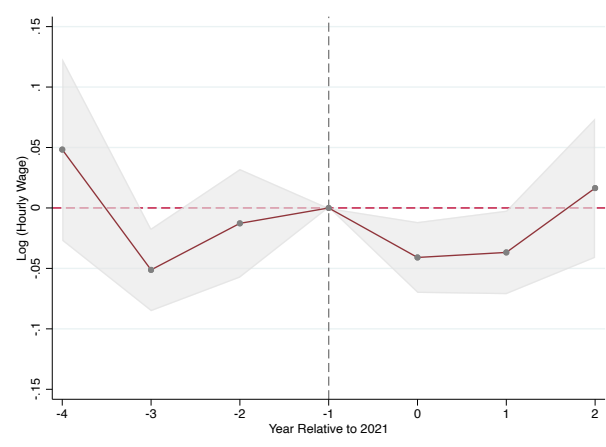
Note: The table is reproduced from [Feng \(2025\)](#), which studies how employers respond to transparency mandates in job postings.

- “X” indicates that the action is prohibited under the law.
- “✓” indicates that the action is allowed under the law.
- Interview Enquiry refers to whether the law prohibits employers from asking about salary history during job interviews.
- Voluntary Disclosure indicates if the law allows job applicants to voluntarily disclose their salary history without prompting.
- Ask and Verify after Offer refers to whether employers are allowed to verify past salary if and only after a job offer has been made.

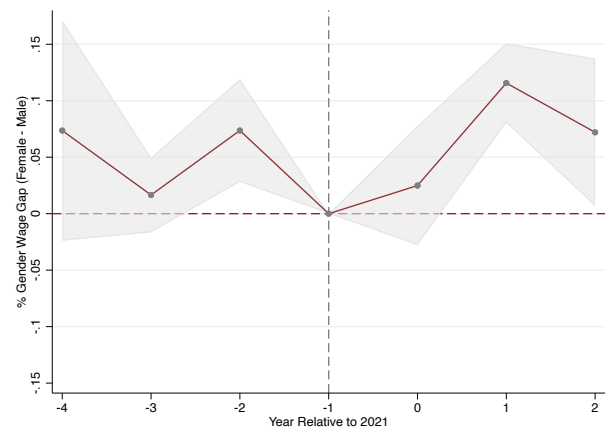
Information on effective dates and the content of the law is available from [here](#).

*California and Washington are excluded from the control group because they both implemented a wage disclosure requirement in 2023. We also exclude New Jersey and New York because both states initially implemented salary history bans only for state employers and agencies, and then expanded them to all employers on 2020-01-01 and 2020-01-06 respectively.

Figure A2: Effect of Pay Transparency Law on Wages by Year



(a) Effect on Wages



(b) Effect on Gender Wage Gap

Note: This figure presents event study estimates of the effects of Colorado’s pay transparency law using data from the Current Population Survey (CPS) Outgoing Rotation Group. The time period is from 2017 to 2023. Panel A shows the point estimates and 95% confidence intervals for the effect of the law on hourly wages estimated with [Equation 12](#), which compares Colorado to a group of control states that implemented salary history bans (SHBs) before 2021 and did not adopt broader pay transparency mandates during 2021–2023. Panel B shows the corresponding estimates for the gender wage gap. These are estimated with [Equation 14](#), which compares wage trends for women and men in Colorado relative to trends in the SHB-only control states. Control variables include age, race, education, calendar month, full-time/part-time status, whether paid hourly, occupation, and industry. Because of the staggered timing of SHB adoption in the control group, we group control states by the year their SHBs took effect, restrict each comparison sample accordingly, and estimate separate event study models for each group. We then construct pooled coefficients for each event time by weighting group-specific estimates according to each group’s share of the combined control sample at that event time. Standard errors are clustered at the Metropolitan Statistical Area (MSA) level.

Table A2: Effect of Pay Transparency Law on Hourly Wage Rates

	Full Sample			Employer Changer			Employer Stayer		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treat	-0.016*	-0.020**	-0.015*	-0.019	0.002	0.015	-0.014*	-0.021**	-0.017*
	(0.009)	(0.009)	(0.009)	(0.033)	(0.032)	(0.031)	(0.008)	(0.009)	(0.009)
Control	X	X	X	X	X	X	X	X	X
State FE	X	X	X	X	X	X	X	X	X
MSA FE	X	X	X	X	X	X	X	X	X
Year FE	X	X	X	X	X	X	X	X	X
Occupation		X	X		X	X		X	X
Industry			X			X			X
Observations	68871	66537	66210	4995	4817	4795	63875	61718	61413
Mean	20.243	20.169	20.169	19.193	19.076	19.070	20.325	20.254	20.254

Note: This table reports the effects of Colorado’s pay transparency law on realized wages, using data from the Current Population Survey (CPS) Outgoing Rotation Group and estimated via specification [Equation 11](#). The dependent variable is the log of the hourly wage rate. The difference-in-differences design compares Colorado to a group of control states that implemented salary history bans (SHBs) before 2021 and did not adopt broader pay transparency mandates between 2021 and 2023. To address staggered SHB adoption across control states, we group control states by the year their SHBs took effect and restrict each comparison sample accordingly. We estimate separate DID models for each group and compute a weighted average treatment effect, where weights correspond to each group’s share of the combined control sample. The CPS ORG records earnings only during respondents’ 4th and 8th survey months. Columns (1)–(3) include the full sample; columns (4)–(6) restrict to workers who changed employers between the 4th and 8th months; columns (7)–(9) restrict to those who did not. All specifications control for age, race, education, worker classification, calendar month, full-/part-time status, and whether paid hourly. Standard errors are clustered at the Metropolitan Statistical Area (MSA) level. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Table A3: Effect of Pay Transparency Law on the Gender Wage Gap

	Full Sample			Subsample: Emp Changer			Subsample: Emp Stayer		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female×Post×CO	0.031* (0.016)	0.028 (0.017)	0.026* (0.015)	0.152** (0.065)	0.141** (0.064)	0.159** (0.064)	0.025 (0.017)	0.022 (0.016)	0.024 (0.016)
Female×CO	-0.011 (0.012)	-0.001 (0.014)	0.002 (0.011)	-0.097* (0.050)	-0.071 (0.050)	-0.088* (0.050)	-0.007 (0.012)	0.003 (0.012)	0.002 (0.012)
Post×CO	-0.029*** (0.011)	-0.033*** (0.012)	-0.029*** (0.010)	-0.079* (0.046)	-0.058 (0.044)	-0.052 (0.043)	-0.026** (0.012)	-0.029*** (0.011)	-0.028** (0.011)
Female×Post	0.001 (0.008)	0.003 (0.008)	0.003 (0.008)	-0.030 (0.031)	-0.018 (0.030)	-0.019 (0.029)	0.003 (0.008)	0.003 (0.008)	0.003 (0.008)
Female	-0.142*** (0.006)	-0.124*** (0.006)	-0.106*** (0.006)	-0.107*** (0.025)	-0.096*** (0.024)	-0.087*** (0.024)	-0.144*** (0.006)	-0.111*** (0.006)	-0.110*** (0.006)
Control	X	X	X	X	X	X	X	X	X
State FE	X	X	X	X	X	X	X	X	X
MSA FE	X	X	X	X	X	X	X	X	X
Year FE	X	X	X	X	X	X	X	X	X
Occupation		X	X		X	X		X	X
Industry			X			X			X
Observations	68871	66537	66210	4995	4817	4795	63875	61718	61413
Mean	20.243	20.169	20.169	19.193	19.076	19.070	20.325	20.254	20.254

Note: This table reports the effects of Colorado’s pay transparency law on the gender wage gap, using data from the Current Population Survey (CPS) Outgoing Rotation Group and estimated via specification [Equation 13](#). The time period is from 2017 to 2023 and the dependent variable is the log of the hourly wage rate. The triple difference design compares Colorado to a group of control states that implemented salary history bans (SHBs) before 2021 and did not adopt broader pay transparency mandates between 2021 and 2023. All coefficients are constructed using a two-step procedure. First, we group control states by the year their salary history bans (SHBs) took effect, restrict each comparison sample to data after the SHB implementation date, and estimate a separate triple-difference model for each group. Second, we compute a pooled estimate for each coefficient by weighting the group-specific estimates according to each group’s share of observations in the combined control sample. This approach applies to all coefficients in the specification. The coefficient on ‘Female×Post×CO’ identifies the differential effect of the law on women relative to men—i.e., the change in the gender wage gap attributable to the law. The coefficient on ‘Post×CO’ captures the post-law effect for men. The coefficient on ‘Female×CO’ reflects pre-law gender wage differences between Colorado and the control states, while ‘Female’ captures the baseline gender wage gap in the control states. Columns (1)–(3) use the full sample; columns (4)–(6) restrict to workers who changed employers between their 4th and 8th CPS survey months; columns (7)–(9) restrict to those who did not. The CPS ORG collects wage data only in months 4 and 8. All regressions control for age, race, education, worker classification, calendar month, full-/part-time status, and whether paid hourly. Standard errors are clustered at the Metropolitan Statistical Area (MSA) level. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Table A4: Effect of Pay Transparency Law on the Gender Wage Gap, Employer Stayer Subsample

	Subsample: Occ Changer			Subsample: Occ Stayer		
	(1)	(2)	(3)	(4)	(5)	(6)
Female×Post×CO	0.013 (0.026)	0.017 (0.025)	0.022 (0.025)	0.039* (0.022)	0.025 (0.020)	0.024 (0.020)
Female×CO	0.002 (0.019)	0.006 (0.018)	0.003 (0.018)	-0.018 (0.017)	0.002 (0.016)	0.002 (0.015)
Post×CO	-0.014 (0.018)	-0.021 (0.018)	-0.019 (0.017)	-0.036** (0.015)	-0.036** (0.014)	-0.033** (0.014)
Female×Post	0.024* (0.012)	0.017 (0.012)	0.018 (0.012)	-0.015 (0.011)	-0.009 (0.011)	-0.010 (0.010)
Female	-0.173*** (0.009)	-0.133*** (0.010)	-0.130*** (0.010)	-0.121*** (0.009)	-0.092*** (0.009)	-0.091*** (0.009)
Control	X	X	X	X	X	X
State FE	X	X	X	X	X	X
MSA FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Occupation		X	X		X	X
Industry			X			X
Observations	27,229	26,089	25,962	36,943	35,915	35,729
Mean	20.583	20.507	20.502	20.158	20.121	20.124

Note: This table reports the effects of Colorado’s pay transparency law on the gender wage gap, using data from the Current Population Survey (CPS) Outgoing Rotation Group and estimated via specification [Equation 13](#). The time period is from 2017 to 2023 and the dependent variable is the log of the hourly wage rate. The sample is restricted to workers who did not change employers during the CPS survey period. The triple difference design compares Colorado to a group of control states that implemented salary history bans (SHBs) before 2021 and did not adopt broader pay transparency mandates between 2021 and 2023. All coefficients are constructed using a two-step procedure. First, we group control states by the year their SHBs took effect, restrict each comparison sample to data after the SHB implementation date, and estimate a separate triple-difference model for each group. Second, we compute a pooled estimate for each coefficient by weighting the group-specific estimates according to each group’s share of observations in the combined control sample. This approach applies to all coefficients in the specification. The coefficient on ‘Female×Post×CO’ identifies the differential effect of the law on women relative to men—i.e., the change in the gender wage gap attributable to the law. The coefficient on ‘Post×CO’ captures the post-law effect for men. The coefficient on ‘Female×CO’ reflects pre-law gender wage differences between Colorado and the control states, while ‘Female’ captures the baseline gender wage gap in the control states. Columns (1)–(3) restrict the sample to employer stayers who changed occupations between their 4th and 8th CPS survey months; columns (4)–(6) restrict to those who did not change occupations. The CPS ORG collects wage data only in months 4 and 8. All regressions control for age, race, education, worker classification, calendar month, full-/part-time status, and whether paid hourly. Standard errors are clustered at the Metropolitan Statistical Area (MSA) level. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Table A5: Effect of Pay Transparency Law on the Gender Wage Gap, by Occupational Gender Composition

	Occupations $\geq 50\%$ Female			Occupations $< 50\%$ Female		
	(1)	(2)	(3)	(4)	(5)	(6)
Female $\times$ Post $\times$ CO	0.041 (0.027)	0.049* (0.026)	0.053** (0.026)	-0.011 (0.026)	-0.019 (0.026)	-0.007 (0.026)
Female	-0.092*** (0.009)	-0.107*** (0.009)	-0.105*** (0.009)	-0.142*** (0.010)	-0.139*** (0.010)	-0.132*** (0.009)
Female $\times$ CO	0.007 (0.021)	0.007 (0.020)	0.004 (0.020)	0.027 (0.021)	0.044** (0.021)	0.040** (0.020)
Post $\times$ CO	-0.035 (0.023)	-0.049** (0.022)	-0.049** (0.022)	-0.032** (0.013)	-0.025* (0.013)	-0.021* (0.013)
Female $\times$ Post	0.004 (0.013)	0.004 (0.012)	0.005 (0.012)	0.012 (0.012)	0.016 (0.012)	0.013 (0.012)
Control	X	X	X	X	X	X
State FE	X	X	X	X	X	X
MSA FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Occupation		X	X		X	X
Industry			X			X
Observations	33,545	31,904	31,773	35,638	34,933	34,727

Note: This table reports estimates from a triple-difference model assessing the impact of Colorado's pay transparency law on the gender wage gap. The outcome variable is the log of hourly wages, using CPS Outgoing Rotation Group data from 2017 to 2023. The sample is limited to workers who remained with the same employer across survey months. The key coefficient of interest, *Female $\times$ Post $\times$ CO*, captures the differential effect of the law on women in Colorado relative to men and to workers in control states. The control group consists of states that implemented salary history bans (SHBs) prior to 2021 and did not introduce additional pay transparency mandates through 2023. Estimates are obtained in two steps: (1) we estimate separate models for each SHB cohort group using data after their SHB effective dates; (2) we then pool estimates, weighting by each group's share of the control sample. Columns (1)–(3) restrict to occupations where at least 50% of workers are female; columns (4)–(6) restrict to occupations with less than 50% female. All models control for age, race, education, full-/part-time status, worker classification, calendar month, and hourly pay indicator. Standard errors are clustered at the Metropolitan Statistical Area (MSA) level. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

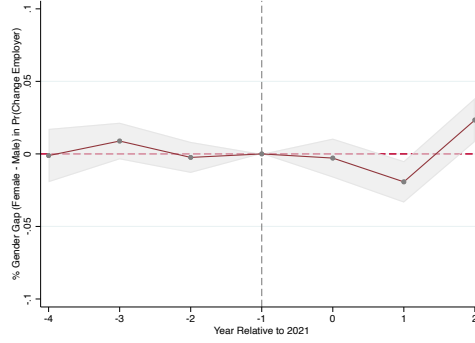
Table A6: Effect of Pay Transparency Law on Gender Gaps in Other Outcomes

	Pr(Change Employer)			Employed		LFP	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female×Post×CO	0.002 (0.003)	0.002 (0.003)	0.002 (0.003)	-0.001 (0.003)	-0.000 (0.003)	0.004 (0.003)	0.005 (0.003)
Female×CO	-0.003 (0.002)	-0.004 (0.002)	-0.003 (0.002)	0.002 (0.002)	0.001 (0.002)	-0.008*** (0.003)	-0.009*** (0.003)
Post×CO	-0.000 (0.002)	0.000 (0.002)	0.001 (0.002)	-0.006*** (0.002)	-0.007*** (0.002)	-0.006*** (0.002)	-0.009*** (0.002)
Female×Post	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)	0.005*** (0.002)	0.004*** (0.002)	-0.001 (0.002)	-0.001 (0.002)
Female	0.000 (0.001)	0.001 (0.001)	0.002 (0.001)	-0.003** (0.001)	-0.004*** (0.001)	-0.012*** (0.001)	-0.013*** (0.001)
Control	X	X	X		X		X
Year FE	X	X	X	X	X	X	X
Occupation		X	X				
Industry			X				
Observations	180,840	174,436	173,521	278,341	278,341	288,613	288,613
Mean	0.020	0.020	0.020	0.980	0.980	0.966	0.966

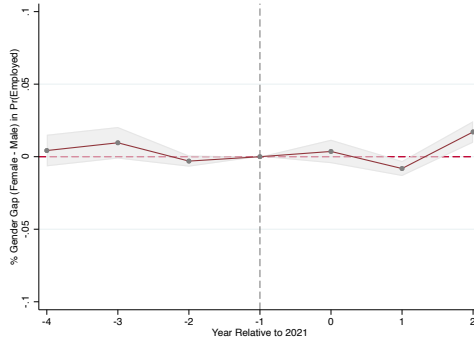
Note: This table reports the effects of Colorado’s pay transparency law on gender gaps in employer changes, employment, and labor force participation, using data from the Current Population Survey (CPS) Basic Monthly files and estimated via specification [Equation 13](#). The time period is from 2017 to 2023. All dependent variables are binary indicators. The triple-difference design compares Colorado to a group of control states that implemented salary history bans (SHBs) before 2021 and did not adopt broader pay transparency mandates between 2021 and 2023. All coefficients are constructed using a two-step procedure. First, we group control states by the year their SHBs took effect, restrict each comparison sample to data after the SHB implementation date, and estimate a separate triple-difference model for each group. Second, we compute a pooled estimate for each coefficient by weighting the group-specific estimates according to each group’s share of observations in the combined control sample. This approach applies to all coefficients in the specification. The coefficient on ‘Female×Post×CO’ identifies the differential effect of the law on women relative to men—i.e., the change in the gender gap in the outcome. The coefficient on ‘Post×CO’ captures the post-law effect for men. The outcome for columns (1) through (3) is an indicator for changing employers, columns (4) through (5) is an indicator for employment, and columns (6) through (7) is an indicator for labor force participation. All columns except (4) and (6) include controls for age and calendar month. Standard errors are clustered at the Metropolitan Statistical Area (MSA) level. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Figure A3: Effect of Pay Transparency on the Gender Gap in Job Mobility, Employment, and Labor Force Participation

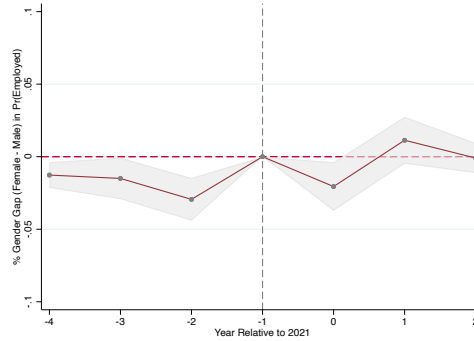
(a) Gender Gap in $\Pr(\text{Change Employer})$



(b) Gender Gap in Employment



(c) Gender Gap in LFP



Note: This figure presents event study estimates of the effects of Colorado’s pay transparency law on gender differences in job mobility, employment, and labor force participation, using data from the Current Population Survey (CPS) Basic Monthly files. The time period is from 2017 to 2023. Each panel shows point estimates and 95% confidence intervals estimated with [Equation 14](#), which compares wage trends for women and men in Colorado relative to trends in the SHB-only control states that implemented salary history bans (SHBs) before 2021 and did not adopt broader pay transparency mandates during 2021–2023. Control variables include age, race, education, and calendar month. Panel (a) shows the effect on the gender gap in the probability of changing employers, Panel (b) in employment, and Panel (c) in labor force participation. Because of the staggered timing of SHB adoption in the control group, we group control states by the year their SHBs took effect, restrict each comparison sample accordingly, and estimate separate event study models for each group. We then construct pooled coefficients for each event time by weighting group-specific estimates according to each group’s share of the combined control sample at that event time. Standard errors are clustered at the Metropolitan Statistical Area (MSA) level.

Table A7: Effect of Pay Transparency Law on Labor Outcomes,
Within-Colorado Comparison

	log(wage)	Gender Wage Gap	Pr(Change Employer)
	(1)	(2)	(3)
Female×Private×Post		0.017 (0.18)	0.006 (0.47)
Female×Private		-0.001 (-0.03)	0.009 (1.73)
Female×Post		0.010 (0.12)	-0.007 (-0.61)
Private×Post	0.006 (0.34)	-0.000 (-0.00)	-0.012* (-1.60)
Private	0.002 (0.08)	0.002 (0.04)	-0.013 (-1.63)
Female	-0.078* (-6.23)	-0.085* (-2.25)	-0.010 (-2.91)
Control	X	X	X
Year	X	X	X
Occupation	X	X	X
Industry	X	X	X
Observations	8176	8176	21429
Adjusted R^2	0.492	0.492	0.014
Mean	21.62	21.62	0.0225

Note: This table shows the effects of the pay transparency law on labor market outcomes using data from the Current Population Survey (CPS) between 2017 and 2023. The outcomes include: log hourly wage (Columns 1 and 2) and the gender gap in the probability of changing employers (Column 3). Estimates in Column 1 are based on the DID specification [Equation 12](#), while Columns 2 and 3 are estimated using the triple-difference model described in [Equation 13](#). For Columns 2 and 3, the coefficient on ‘Private×Post’ denotes the effect on men, and the coefficient on ‘Female×Private×Post’ denotes the effect on the gender gap. The empirical design is a within-Colorado DID/triple-difference approach that compares workers in the private sector (treated group) to those in the public sector (control group), before and after the implementation of the law in 2021. Columns 1 and 2 control for age, race, education, calendar month, full-time/part-time status, and whether paid hourly. Column 3 controls for age, race, education, and calendar month. Wild cluster bootstrap t-statistics are in parentheses, clustered at the metropolitan area (MSA) level, with 1999 replications. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Table A8: Effect of Pay Transparency Law on Wages, CO vs. Unregulated States

	log(hourly wage)			Gender Wage Gap		
	Full Sample	Emp Changer	Emp Stayer	Full Sample	Emp Changer	Emp Stayer
	(1)	(2)	(3)	(4)	(5)	(6)
Female×Post×CO				0.018 (0.012)	0.092*** (0.029)	0.015 (0.013)
Female×CO				0.015 (0.007)	0.048 (0.026)	0.013 (0.007)
Female×Post				0.008 (0.004)	-0.012 (0.012)	0.009 (0.005)
Post×CO	0.004 (0.006)	-0.002 (0.020)	0.004 (0.005)	-0.004 (0.009)	-0.038 (0.024)	-0.002 (0.010)
Female	-0.113*** (0.004)	-0.117*** (0.011)	-0.112*** (0.004)	-0.117*** (0.004)	-0.118*** (0.012)	-0.117*** (0.004)
Control	X	X	X	X	X	X
State FE	X	X	X	X	X	X
MSA FE	X	X	X	X	X	X
Occupation	X	X	X	X	X	X
Industry	X	X	X	X	X	X
Observations	134218	9188	124932	134218	9188	124932
Adjusted R^2	0.483	0.510	0.482	0.483	0.512	0.482
Mean	18.754	17.556	18.847	18.754	17.556	18.847

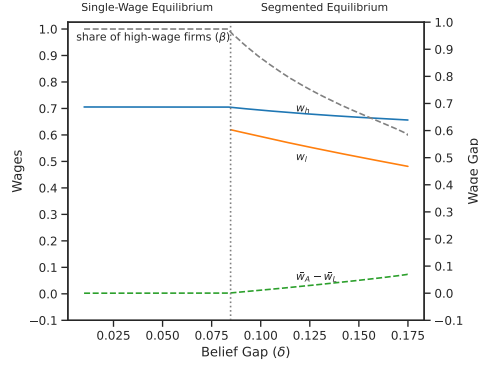
Note: This table reports the effects of Colorado's pay transparency law on hourly wages and gender wage gaps using data from the Current Population Survey (CPS) from 2017 to 2023. Estimates in Columns 1-3 are based on the DID specification [Equation 12](#), while Columns 4-6 are estimated using the triple-difference model described in [Equation 13](#). The empirical design compares labor market outcomes in Colorado to a group of unregulated states that did not adopt salary history bans or pay transparency mandates during this period. The outcome is the log hourly wage. Columns (1), (4) use the full sample, Columns (2), (5) restrict to workers who changed employers between the 4th and 8th CPS survey months, and Columns (3), (6) restrict to workers who stayed with the same employer. All regressions control for age, race, education, calendar month, full-time/part-time status, and whether paid hourly. In addition, all models include state fixed effects, metropolitan statistical area (MSA) fixed effects, occupation fixed effects, and industry fixed effects. Standard errors are clustered at the metropolitan area (MSA) level. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Table A9: Effect of Pay Transparency Law on Other Labor Outcomes, CO vs. Unregulated States

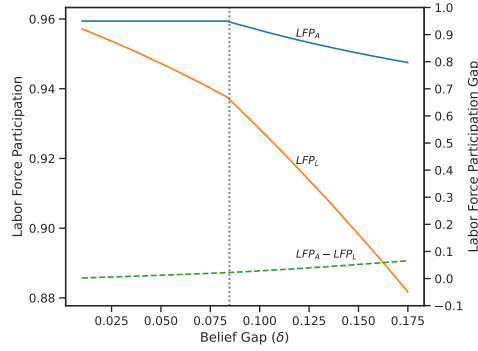
	Pr(Change Employer)	Employed	LFP
	(1)	(2)	(3)
Female×Post×CO	-0.003 (0.003)	-0.001 (0.004)	0.000 (0.006)
Female×CO	-0.002 (0.002)	0.003 (0.002)	0.001 (0.002)
Post×CO	0.000 (0.003)	-0.004 (0.002)	-0.002** (0.001)
Female×Post	0.001 (0.001)	0.001 (0.001)	0.001 (0.002)
Female	-0.000 (0.001)	-0.003*** (0.001)	-0.022*** (0.002)
Control	X	X	X
Year	X	X	X
Occupation	X		
Industry	X		
Observations	345886	557266	578269
Adjusted R^2	0.005	0.011	0.015
Mean	0.020	0.982	0.964

Note: This table reports the effects of Colorado’s pay transparency law on other labor market outcomes using data from the Current Population Survey (CPS) from 2017 to 2023. The estimates are based on a triple-difference specification (see `autorefgenderdid`) that compares Colorado’s outcomes to those of a group of unregulated states that did not adopt salary history bans or pay transparency mandates during this period. The outcomes are: (1) a binary indicator of changing employers, (2) employment, and (3) labor force participation. Column (1) includes controls for age, race, education, calendar month, occupation, and industry. Columns (2) and (3) control for age, race, education, and calendar month. All specifications include state and metropolitan statistical area (MSA) fixed effects. The coefficient on ‘Female×Post×CO’ reflects the change in the gender gap attributable to the law. The coefficient on ‘Post×CO’ captures the effect of the law on men in Colorado relative to men in the control states. Standard errors are clustered at the MSA level. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

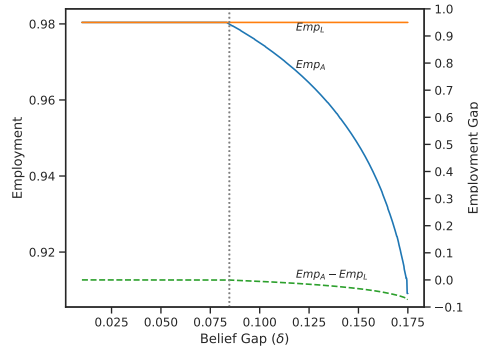
Figure B4: Labor Market Equilibrium and the Gender Belief Gap



(a) Wage



(b) Labor Force Participation



(c) Employment

Note: Panel A plots equilibrium wages and the share of high-wage firms, β , as a function of the degree of the gender belief gap, δ . Panel B and Panel C plot labor force participation and employment rates as functions of δ , respectively. The dotted vertical line marks the cutoff value of the gender belief gap that induces a switch from a single to a segmented labor market. We use the following parameter values: $\alpha = 0.5$, $\eta = 0.7$, $s = 0.02$, $\mu = 0.02$, $\sigma^2 = 2.5$, and $N/L = 1$.

C Segregation Across Employers: LinkedIn Data and Estimates

C.1 Sample

Employment spells for 2017–2022 come from LinkedIn profiles ([People Data Labs, 2022](#)), restricted to Colorado and the eight comparison states. A spell records the employer, the job title, the employer’s industry and size band, and the start and end dates. A worker’s state is their own recorded location, not the employer’s headquarters. The person-year panel covers 2018–2022 and includes a worker in year t if they held a job on 31 December of year $t - 1$, giving 21.9 million person-years. Two consecutive spells at different employers count as an employer change if the second begins no more than one month before the first ends and no more than one year after it. A spell is defined by employer and title jointly, so a new spell at the same employer is a title change.

C.2 Measures

Each employer is assigned the share of women among the distinct workers it employed before 2021, and this share is held fixed for all later years. When the share is used as the outcome for a given worker, that worker is left out of the calculation. Employers with fewer than ten observed workers are dropped. The remaining employers account for 76.6 percent of person-years and form the sample for [Table C10](#) and [Table C12](#); [Table C11](#) uses the full panel.

C.3 Results

All estimates are pooled nine-state triple differences with every lower-order term and standard errors clustered by state.⁷ [Table C10](#) absorbs state and year, and is discussed in the text. Column 2 adds industry and the estimate falls to -0.0002 : within an industry, women in Colorado worked for the same employers after the law as before, and the little movement there is occurs across industries.

[Table C11](#) uses the full panel and absorbs state, year, lagged industry and lagged employer-size category. The estimate is -0.0002 for changing employer, against a base rate

⁷These estimates depart from [Equation 13](#) in four ways. They pool the nine states over 2017–2022 rather than grouping the comparison states by the year their salary history bans took effect and weighting the group estimates by observation share. They include none of the individual controls: the profiles record no race, education, hours, or pay basis, and job titles are not mapped to occupation codes. The sample is not restricted to workers aged 22 to 64, because a birth year is recorded for only about 20 percent of profiles. Standard errors are clustered by state rather than by metropolitan area.

of 8.2 percent, and nearly zero for a title change at the same employer, against 4.7 percent. Neither is significant. This implies that, after the law, women in Colorado moved between employers no more often than men or than women in the comparison states, and changed jobs inside their own employer no more often either. This is what the CPS finds for job mobility.

Table C12 repeats the two outcomes of Table C11 on the restricted sample of Table C10 and, in columns 2 and 4, replaces industry and employer size with employer fixed effects, so that women in Colorado are compared with women at the same employer in the comparison states. The estimates are -0.0009 for changing employer and 0.0003 for a title change. The first is significant on 16.8 million person-years but amounts to one percent of the base rate, and is the same size with and without the fixed effects, so it comes from restricting the sample to larger employers rather than from the fixed effects.

Table C10: Segregation Across Employers

	(1) All person-years	(2) Within industry
Female	0.1488*** (0.0036)	0.0571*** (0.0016)
Female $\times$ Post	-0.0049*** (0.0002)	-0.0045*** (0.0001)
Female $\times$ CO	0.0063 (0.0036)	-0.0023 (0.0016)
Post $\times$ CO	0.0000 (0.0003)	-0.0000 (0.0003)
Female $\times$ Post $\times$ CO	-0.0008*** (0.0002)	-0.0002* (0.0001)
Industry fixed effects	No	Yes
Mean of dep. var.	0.4969	0.4969
SD of dep. var.	0.2011	0.2011
Person-years	1.68e+07	1.68e+07

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The dependent variable is the female share of the worker's own employer, measured before 2021 and held fixed, excluding the worker. The coefficient on *Female* is the variance-ratio index of segregation across employers. Column 2 absorbs industry, isolating segregation across employers within an industry. Standard errors are clustered by state.

Table C11: Employer Changes and Within-Employer Title Changes

	(1) Changed employer	(2) Stayed, new job title
Female	0.0014 (0.0011)	0.0042*** (0.0011)
Female $\times$ Post	0.0018*** (0.0003)	0.0018*** (0.0003)
Female $\times$ CO	0.0007 (0.0015)	-0.0003 (0.0011)
Post $\times$ CO	-0.0004 (0.0009)	0.0005 (0.0010)
Female $\times$ Post $\times$ CO	-0.0002 (0.0003)	0.0000 (0.0003)
Mean of dep. var.	0.0815	0.0472
Person-years	2.19e+07	2.01e+07

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Column 2 is restricted to workers who did not change employers. Regressions absorb state, year, lagged industry, and lagged employer-size categories. Standard errors are clustered by state.

Table C12: Employer Changes and Title Changes, Comparing Workers at the Same Employer

	Changed employer		Stayed, new job title	
	(1)	(2)	(3)	(4)
Female	-0.0011 (0.0011)	-0.0004 (0.0013)	0.0042*** (0.0011)	0.0058*** (0.0012)
Female $\times$ Post	0.0007** (0.0003)	0.0001 (0.0002)	0.0017*** (0.0002)	0.0018*** (0.0002)
Female $\times$ CO	0.0007 (0.0015)	0.0017 (0.0017)	-0.0005 (0.0012)	-0.0001 (0.0012)
Post $\times$ CO	0.0002 (0.0008)	0.0018 (0.0011)	-0.0000 (0.0010)	-0.0008 (0.0008)
Female $\times$ Post $\times$ CO	-0.0007** (0.0003)	-0.0009*** (0.0002)	0.0002 (0.0002)	0.0003 (0.0002)
Employer fixed effects	No	Yes	No	Yes
Mean of dep. var.	0.0815	0.0815	0.0540	0.0540
Person-years	1.68e+07	1.68e+07	1.68e+07	1.68e+07

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The sample is restricted to workers at employers with at least ten observed workers. Columns 2 and 4 add employer fixed effects, so that women in Colorado are compared with women at the same employer in the comparison states; identification of the triple interaction then comes from employers present in both. Standard errors are clustered by state.